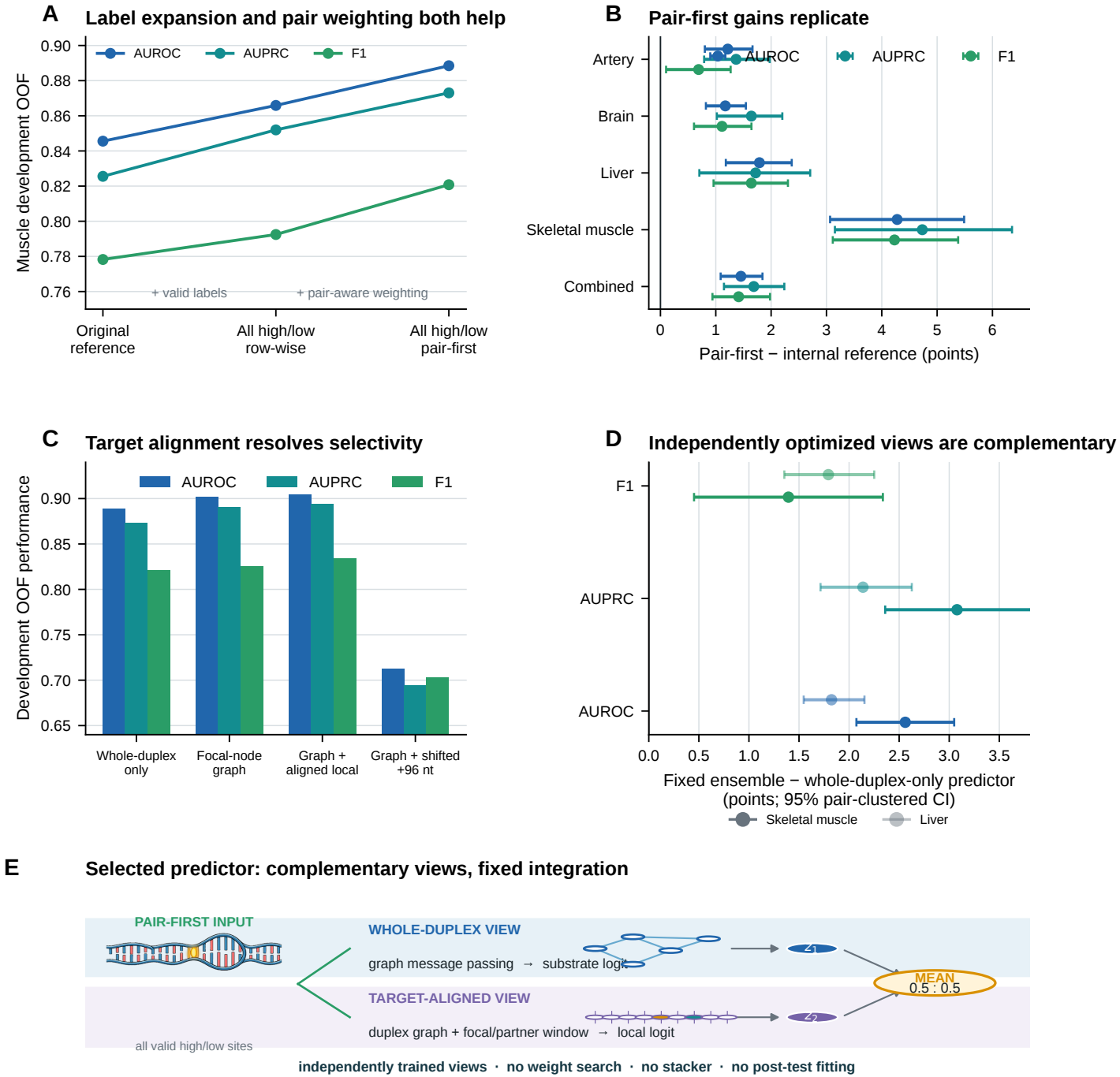

A
Label expansion and pair weighting both help
Muscle development OOF
0.90
0.88
0.86
0.84
0.82
0.80
0.78
0.76
AUROC
AUPRC
F1
+ valid labels
+ pair-aware weighting
Original reference
All high/low row-wise
All high/low pair-first
B
Pair-first gains replicate
Artery
Brain
Liver
Skeletal muscle
Combined
AUROC
AUPRC
F1
0
1
2
3
4
5
6
Pair-first − internal reference (points)
C
Target alignment resolves selectivity
Development OOF performance
0.90
0.85
0.80
0.75
0.70
0.65
AUROC
AUPRC
F1
Whole-duplex only
Focal-node graph
Graph + aligned local
Graph + shifted +96 nt
D
Independently optimized views are complementary
F1
AUPRC
AUROC
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
Fixed ensemble − whole-duplex-only predictor
(points; 95% pair-clustered CI)
Skeletal muscle
Liver
E
Selected predictor: complementary views, fixed integration
PAIR-FIRST INPUT
all valid high/low sites
WHOLE-DUPLEX VIEW
graph message passing → substrate logit
TARGET-ALIGNED VIEW
duplex graph + focal/partner window → local logit
z1
z2
MEAN
0.5 : 0.5
independently trained views · no weight search · no stacker · no post-test fitting